When Does Text Help?

A Leakage-Free Information-Richness and Encoder-Adaptation Study for Multimodal Land-Cover Composition Regression

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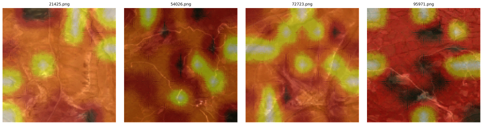
Problem

Predict the **land-cover composition** of a satellite tile:

- 7 mutually exclusive classes (Tree, Shrub, Grass, Crop, Built-up, Barren, Water)
- Percentages sum to 100 %
- **Regression**, not classification

Hypothesis: text captions describing the tile could help a vision model. **Question:** *Does adding text actually help — and can we measure it honestly?*



Cross-attention overlays on ARAS400k tiles

Dataset & Phase 1 (PoC)

ARAS400k: 10 000 RGB tiles (256×256), pixel masks, **5 caption variants**.

Architecture (proof-of-concept):

$$\underbrace{\text{Image}}_{\text{CLIP ViT-B/32}} \rightarrow \underbrace{\text{49 patch tokens}}_{Q} + \underbrace{\text{77 text tokens}}_{K,\,V}$$

$$\rightarrow \text{ 8-head cross-attention } \rightarrow \text{ mean-pool } \rightarrow \text{ MLP } \rightarrow 7\,\% \text{ (softmax)}$$

Loss: MSE on softmax output vs. GT/100.

Phase 1 was a *single-sample forward pass* — pipeline validation only, no training yet.

The Twist: Data Leakage

3 of 5 caption variants embed GT % verbatim

`hybrid_gemma3-4b`	*"grasslands (92%) with sparse barren land (6%)"*
`hybrid_qwen3-vl-8b`	*"grassland covering 92% of the area"*
`text_qwen3-4b`	*"predominantly covered by grass..."*

A model trained on these solves the task **lexically** — it reads the answer.

Crucially: stripping digits is not enough. The VLMs that wrote those captions *saw* the GT, so even qualitative narrative carries label structure. The leak is **structural, not lexical**.

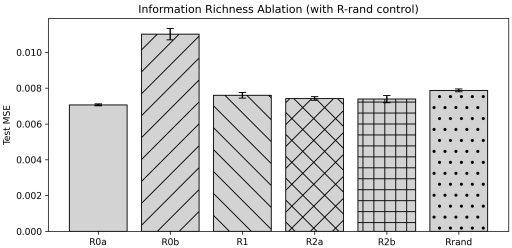
Redesign: use only the two `vision_*` captions (image-only generation), plus regex sanitization as a safety filter.

Phase 2: Information Richness Ablation

Code	Architecture	Text input
R0a	CLIP + cross-attn + MLP	`""` empty token
R0b	CLIP + GAP + MLP	— (no text path)
R1	+ cross-attn + MLP	noun-lemma keywords over both `vision_*`
R2a	+ cross-attn + MLP	sanitized `vision_gemma3-4b`
R2b	+ cross-attn + MLP	sanitized `vision_qwen3-vl-8b`
R-rand	+ cross-attn + MLP	77 random vocab tokens (control)

- **6 conditions** $\times$ 3 seeds = **18 runs**
- CLIP **frozen**; only fusion + MLP train ($\sim$1.18 M params)
- Precompute trick $\Rightarrow$ full grid $<$ **50 min on M4**

Phase 2 Results — a Surprise



Test MSE, 6 conditions (mean $\pm$ std, 3 seeds)

Condition	**Test MSE** $\downarrow$
R0a (empty)	**0.00707**
R2b (qualitative)	0.00739
R2a (qualitative)	0.00751
R1 (keywords)	0.00760
R-rand	0.00787
R0b (no text)	**0.01101**

Surprises:

- Empty caption *beats* every richer one.
- Largest gap is R0a vs. R0b $\Rightarrow$ **architecture**, not text, drives the win.

Why? Two Follow-up Controls

Caption-fidelity stratified test:
tag each test tile `agree`/`disagree` by whether the caption mentions the dominant GT class.

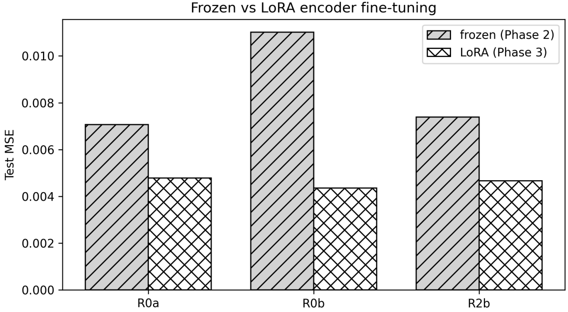
	MSE (agree)	MSE (disagree)
R0a	0.00676	0.00740
R2a	**0.00671**	0.00823
R2b	0.00680	0.00787

Fidelity, not richness, gates the effect. Wins and losses cancel on average.

Random-text control (R-rand):
77 random CLIP tokens, fixed seed.

- R0a – R-rand:
  $p = 8.6 \times 10^{-9}$
  $\Rightarrow$ empty token *is* signal
- R1 $\approx$ R-rand ($p = 0.13$)
  $\Rightarrow$ keywords add nothing
- R2b $>$ R-rand ($p = 0.011$)
  $\Rightarrow$ small real signal

Phase 3: LoRA Encoder Adaptation (severe change)



	Frozen	LoRA	Δ
R0a	0.00707	0.00478	−32 %
R0b	0.01101	**0.00435**	−60 %
R2b	0.00739	0.00466	−37 %

- LoRA (r=8, α=16, last 6 blocks)
- Only MLP projections adapted (attention MHA bypasses module forward)
- Gradient clipping (max-norm 1.0) fixed R0a divergence
- Online training, ∼84 s/epoch on M4

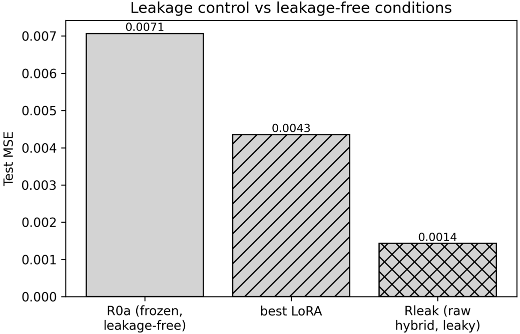
Regime Reversal + Leakage Quantified

Regime reversal:

Frozen: R0a (0.0071) $\ll$ R0b (0.0110)
cross-attn **helps**

LoRA: R0b (0.0044) $<$ R2b (0.0047) $<$ R0a (0.0048
pure-vision **wins**;
fusion now redundant

Once the encoder can adapt, the fusion layer's capacity contribution becomes unnecessary — the visual pathway dominates.



Rleak (raw hybrid, GT % intact) $=$ **0.00143**
$\approx$ **1/3** of best leakage-free model
$\approx$ **1/5** of frozen R0a

The model copies the answer from text.
Leakage failure mode **quantified**, not just asserted.

Code Base & Reproducibility

```
src/
   dataset.py      7 conditions, 70/15/15
   sanitize.py     regex + spaCy + fidelity
   model.py        2 regressors + freeze flag
   train.py        AdamW, cosine, W&B, LoRA
   eval.py         metrics + attention viz
   precompute.py   CLIP cache (image+text)
   lora.py         LoRALinear + inject_lora

notebooks/
   phase-1/phase1_poc.ipynb
   phase-2/phase2_main.ipynb
   phase-3/phase3_main.ipynb
```

Reproducibility:

- `requirements.txt`
- Deterministic 70/15/15 (seed 42)
- 3 training seeds: 42, 1337, 2024
- Heavy artefacts gitignored

Tracking:

`wandb.ai/mereyomerfaruk/`
`di725-project` — 18 runs

Switching to notebook `phase3_main.ipynb`

Will show, in order:

- **§2 Rleak training** — per-epoch loss converging
- **§4 evaluation table** — mean $\pm$ std across seeds
- **§5 frozen-vs-LoRA bar chart** — live render
- **W&B project page** — 18 runs, validation curves

Per spec §4.4.1: "explain the code base, how to run it, and demonstrate how it runs."

Conclusions

1. **Fidelity, not richness, gates the effect.**
   Clean captions help *only* when they correctly describe the image; wins and losses cancel on average.
2. **Architecture vs. content is regime-dependent.**
   Frozen: cross-attention drives the gain. LoRA: pure-vision wins; fusion redundant.
3. **Leakage quantified, not just asserted.**
   Rleak $\approx 1/5$ of frozen R0a — the failure mode is real and measurable.
4. **Methodological takeaway:**
   multimodal gains must be ablated against architecture-matched no-text, random-text, *and* fidelity-stratified baselines before any non-trivial text contribution is claimed.

Thank you.

github.com/OmerFarukMerey/remote-sensing-caption-composition

wandb.ai/mereyomerfaruk/di725-project